Electrical and Computer Engineering
University of Thessaly

**ECE327 - Digital Systems VLSI**

# CMOS Logic Design, Layout, and Delay Analysis

Winter Semester — 2022-2023

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# 1 Schematic Design and Transistor Sizing

The implementation of the boolean function $f_{aoi21} = (a + bc)' = a'(b' + c')$ is shown in figure 1.

As established in VLSI theory, the channel width ($W$) of a transistor is inversely proportional to its equivalent resistance ($R_{eq}$). Consequently, by applying the inverse rule of three, we determine the transistor widths for each conduction path. This ensures that the total resistance for all paths in both the Pull-Up Network (PUN) and the Pull-Down Network (PDN) is equal to the design constraint of $6.5\,k\Omega$.

For instance, in the previously illustrated schematic, the Pull-Up Network (PUN) consists of two primary conduction paths ($ba$ and $ca$), each containing two PMOS transistors in series. To satisfy the design requirements, the widths of these PMOS transistors must be calculated such that the sum of their resistances equals $6.5\,k\Omega$ (assigning $3.25\,k\Omega$ to each transistor).

By applying the inverse proportion rule, we obtain the following:

- A PMOS transistor with width $W = 2\lambda$ exhibits a resistance of $31\,k\Omega$.
- A PMOS transistor with width $W = x$ must exhibit a resistance of $3.25\,k\Omega$.

The calculation is as follows:

$$x = \frac{2\lambda \cdot 31\,k\Omega}{3.25\,k\Omega} \approx 19.07\lambda \tag{1}$$

Therefore, we select a transistor width of $W_p \approx 19\lambda$ for these specific paths to achieve the target resistance.

Consequently, for both conduction paths within the Pull-Up Network, the width of each PMOS transistor is determined to be $19\lambda$.

Regarding the Pull-Down Network (PDN), we analyze the paths ($a$ and $bc$). Similarly, the transistor widths must be calculated so that the total resistance of each path equals $6.5\,k\Omega$. Applying the inverse proportion rule for the $bc$ path (two NMOS transistors in series):

- An NMOS transistor with width $W = 2\lambda$ exhibits a resistance of $13\,k\Omega$.
- An NMOS transistor with width $W = x$ must exhibit a resistance of $3.25\,k\Omega$.

The calculation yields:

$$x = \frac{2\lambda \cdot 13\,k\Omega}{3.25\,k\Omega} = 8\lambda \tag{2}$$

For path $a$, which consists of a single NMOS transistor, the width required to achieve the target resistance of $6.5\,k\Omega$ is calculated as:

$$W_a = \frac{2\lambda \cdot 13\,k\Omega}{6.5\,k\Omega} = 4\lambda \tag{3}$$

Thus, the NMOS transistors in the series path are sized at $W = 8\lambda$, while the single-transistor path uses $W = 4\lambda$.

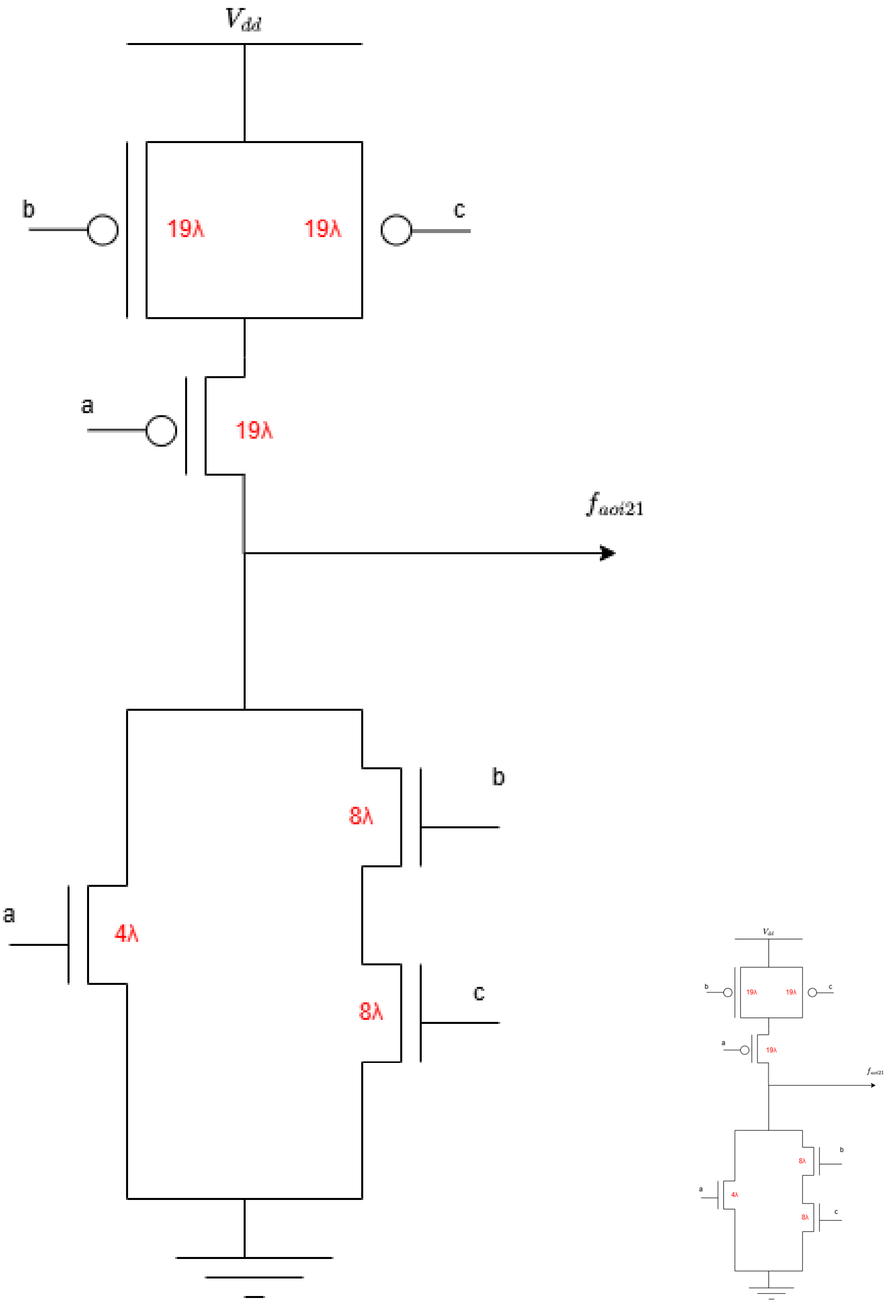


Figure 1: $f_{aoi21}$ function implementation.

The implementation schematics for the remaining Boolean functions are presented below, along with the calculated transistor sizes for their respective Pull-Up (PUN) and Pull-Down Networks (PDN). The size of the transistor for these gates is derived following the exact methodology demonstrated in the previous example, ensuring that each conduction path adheres to the resistance constraint $6.5\,k\Omega$.

The construction of the function $f_{oai31} = (ab + b(c+d))'$, using a minimum number of transistors, is shown in Figure 2. Similarly, the design for the majority function $f_{maj} = (a(b + c) + bc)'$ is illustrated in Figure 3. Lastly, the circuit describing the behavior of the function $f_{aoi22} = (ac + bd)'$ is presented in Figure 4.

In order to describe and effectively design both the Pull-Up (PUN) and Pull-Down (PDN) networks, we utilize the rules of Boolean algebra to rewrite the target functions as follows:

$$f_{oai31} = (ab + bc + bd)' = (b(a + c + d))' = b' + a'c'd' \tag{4}$$

$$f_{maj} = (ab + bc + ac)' = (a(b + c) + bc)' = (a' + b'c') \cdot (b' + c') \tag{5}$$

$$f_{aoi21} = (a + bc)' = a' \cdot (b' + c') \tag{6}$$

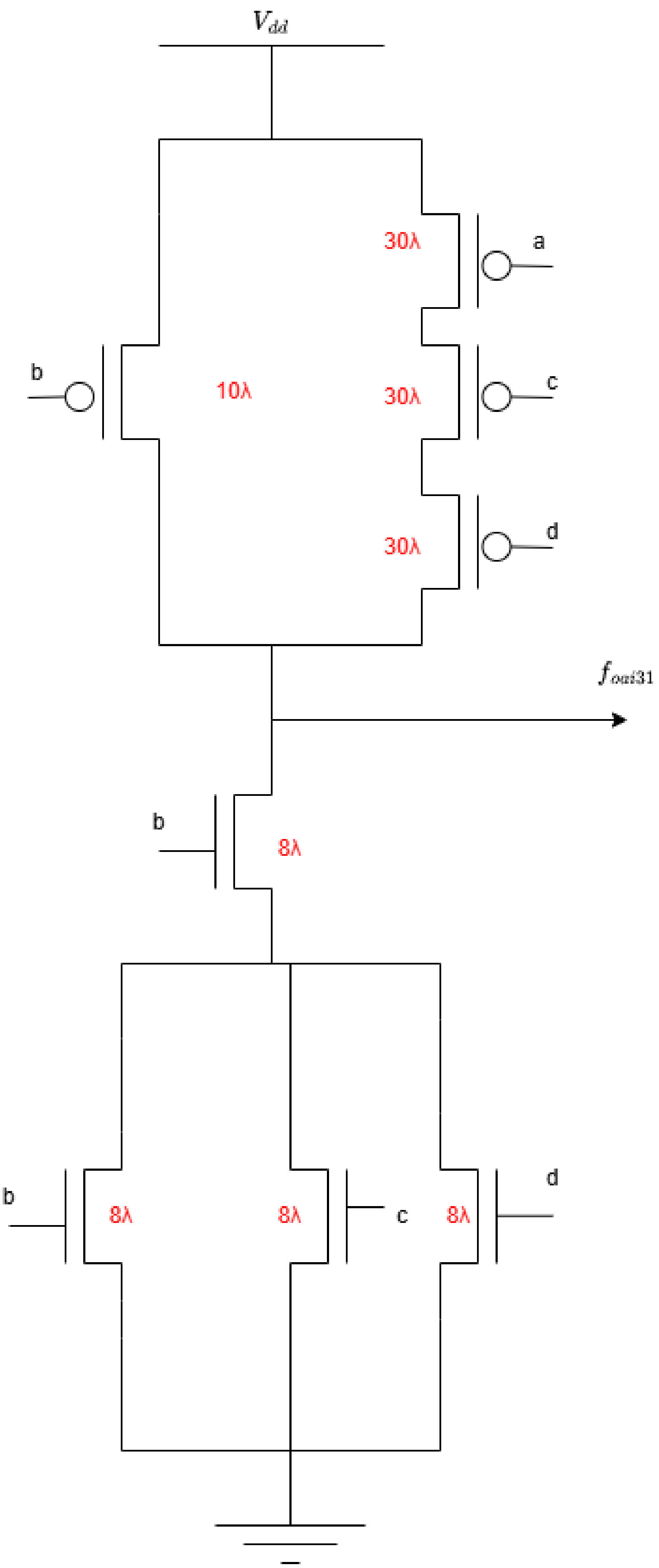


Figure 2: Implementation of the $f_{oai31}$ function.

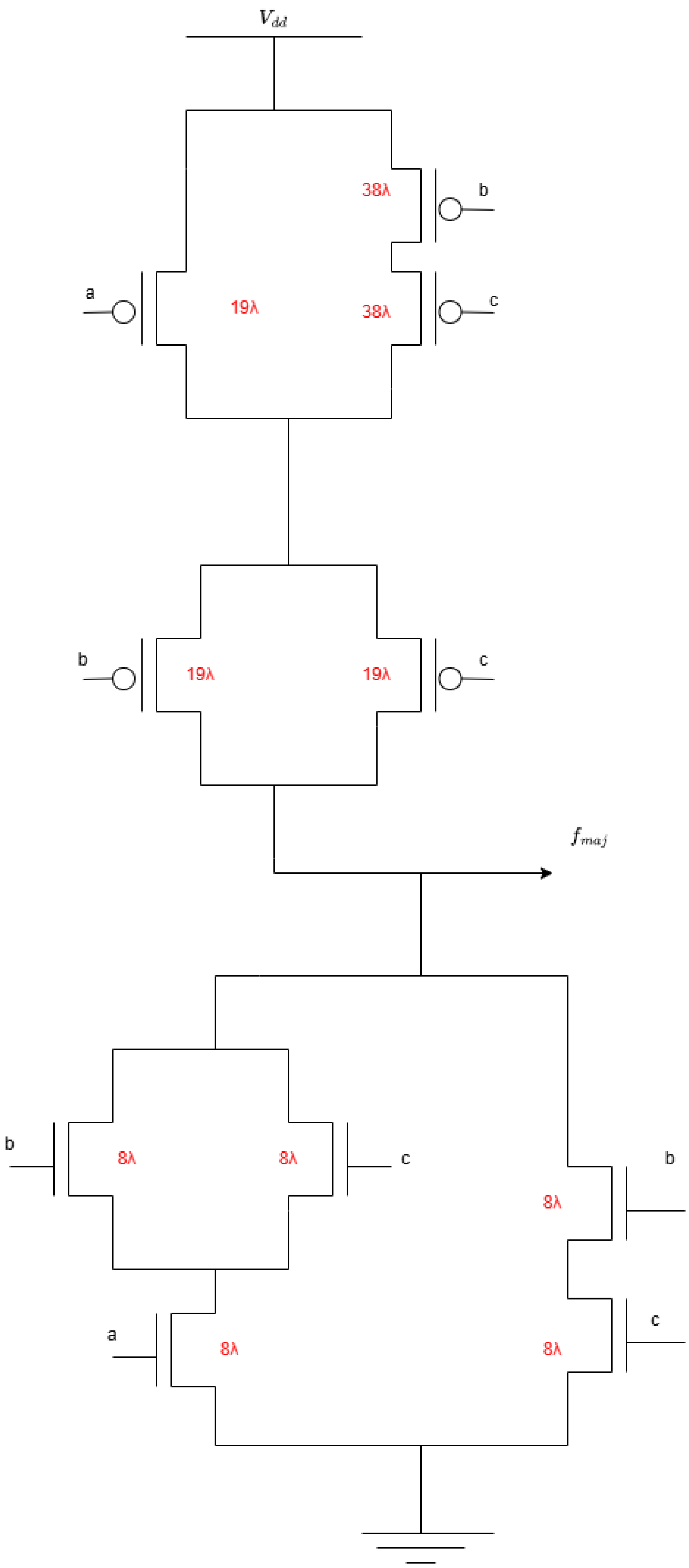


Figure 3: Implementation of the $f_{maj}$ function.

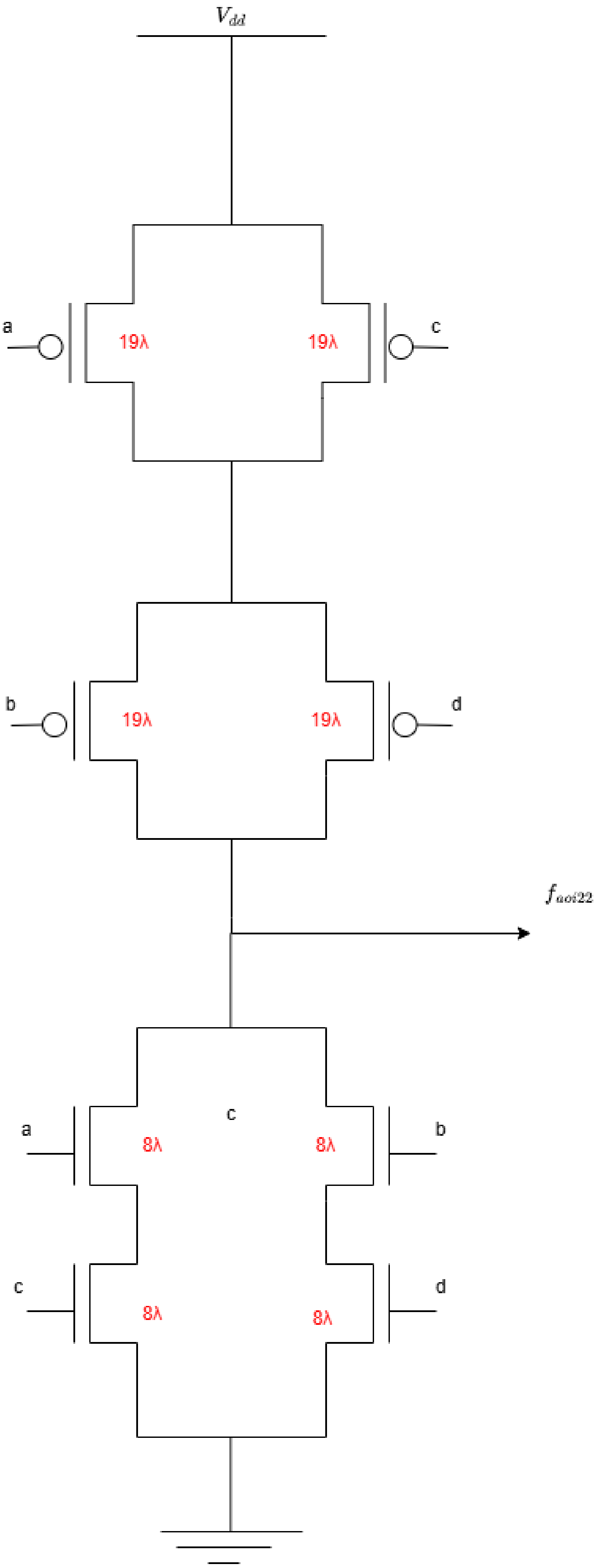


Figure 4: Implementation of the $f_{aoi22}$ function.

# 2 Stick Diagrams

For each schematic implementation developed in the previous section, we aim to design the corresponding dimensionless stick diagrams. To achieve an optimized layout with minimum area and continuous diffusion layers, we identify a common path that traverses all edges of both the Pull-Up (PUN) and Pull-Down (PDN) networks exactly once.

This process, known as finding a common Euler path, allows for the alignment of polysilicon gate strips and minimizes the need for diffusion gaps. Following the identification of these paths, the respective stick diagrams are constructed to demonstrate the placement of transistors and their interconnections.

For the implementation of the binary function $f_{aoi21} = (a + bc)'$, the sequence $A - B - C$ is selected as the common Euler path. The corresponding stick diagram for this path is presented in Figure 5:

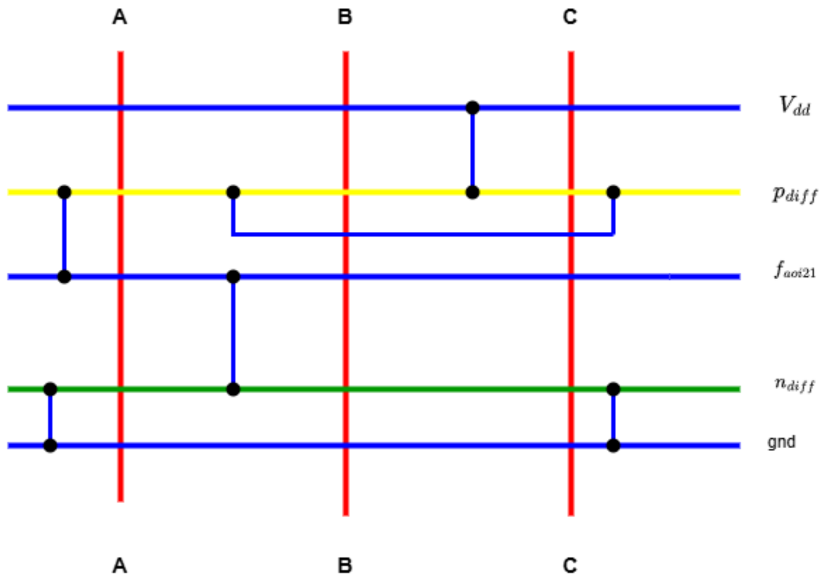


Figure 5: Corresponding stick diagram of the $f_{aoi21}$ function using the $A - B - C$ Euler path.

Similarly for the binary function $f_{oai31} = (ab + b(c + d))'$ ,the sequence $B - A - C - D$ was selected as the common Euler path.The stick diagram for this function is illustrated in figure 6

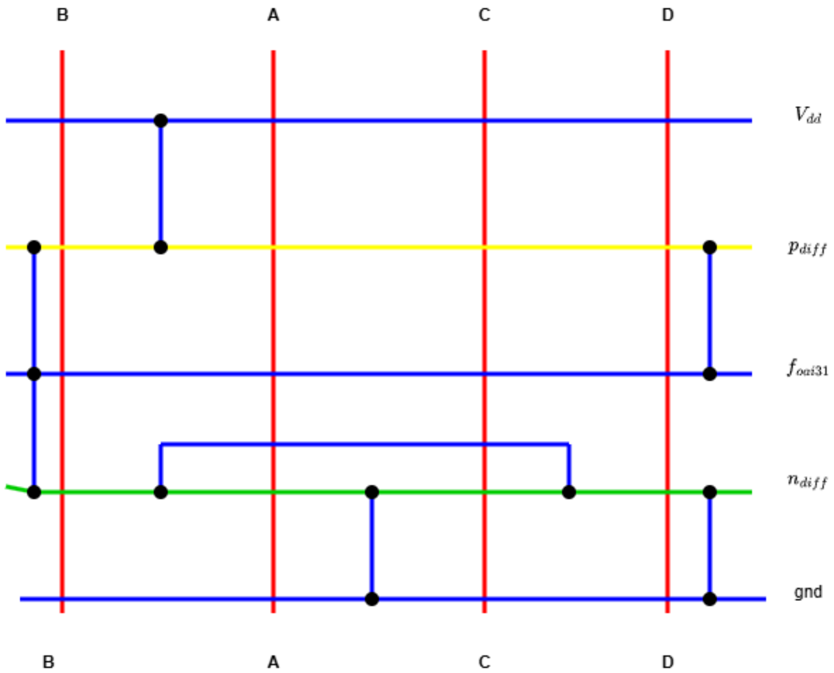


Figure 6: Corresponding stick diagram of the $f_{oai31}$ function using the $B - A - C - D$ Euler path.

Following the same methodology, the stick diagram for the $f_{maj}$ function is constructed. The sequence $C - B - A - C - B$ is designated as the common Euler path to ensure optimal layout connectivity. The resulting stick diagram, which illustrates the transistor placements and the routing of the metal and polysilicon layers, is presented in Figure 7:

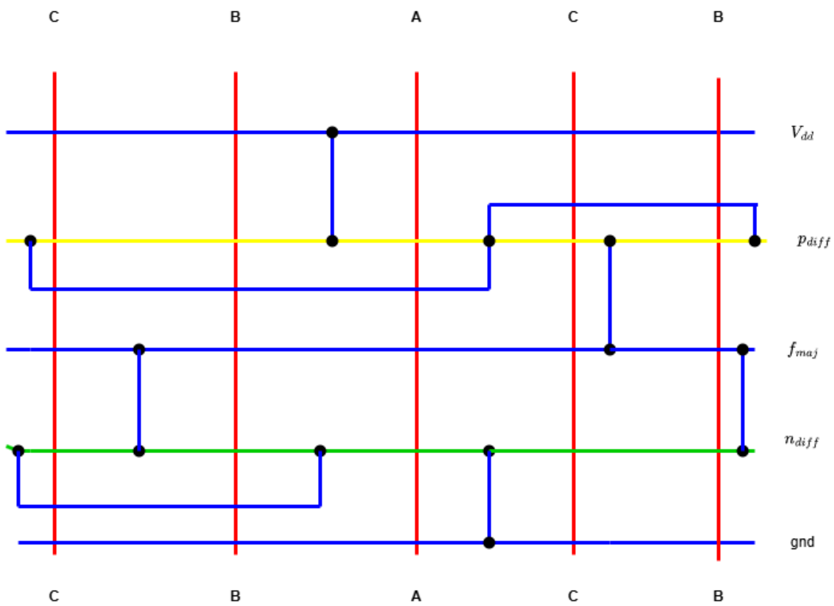


Figure 7: Corresponding stick diagram of the $f_{maj}$ function using the $C - B - A - C - B$ Euler path.

Lastly, for the $f_{aoi22}$ function, the $A - C - D - B$ sequence is selected as the common Euler path. The stick diagram describing this implementation is shown in Figure 8.

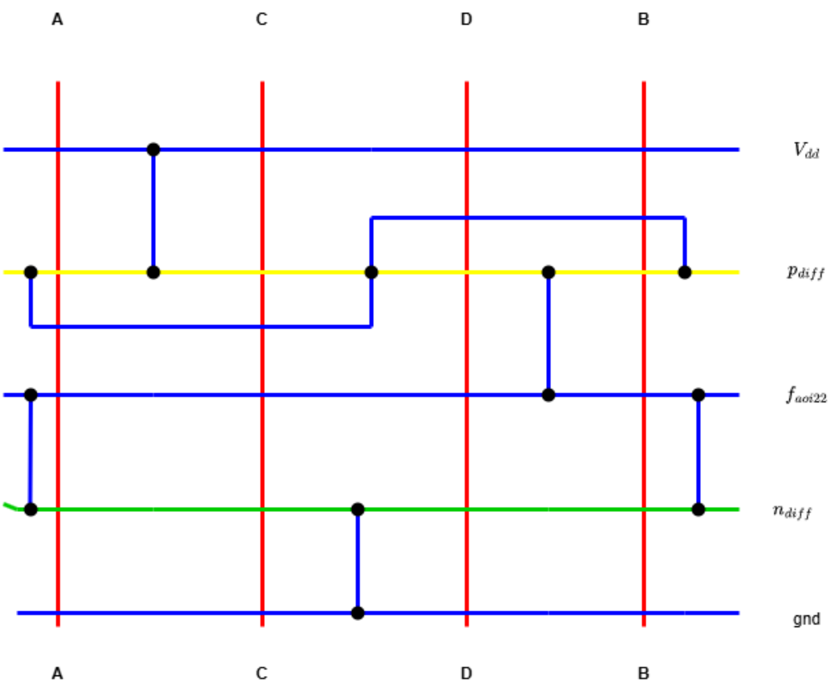


Figure 8: Corresponding stick diagram of the $f_{aoi22}$ function using the $A - C - D - B$ Euler path.

# 3 Layout Design in Magic VLSI and Functionality Verification

In this stage, the previously designed stick diagrams are implemented as standard cells using the Magic VLSI layout tool.The parameter library used for the simulations was EDA TSCMOS 0.25. For each Boolean function, the corresponding physical layout is presented in the following figures (9,**??**,**??**,**??**).

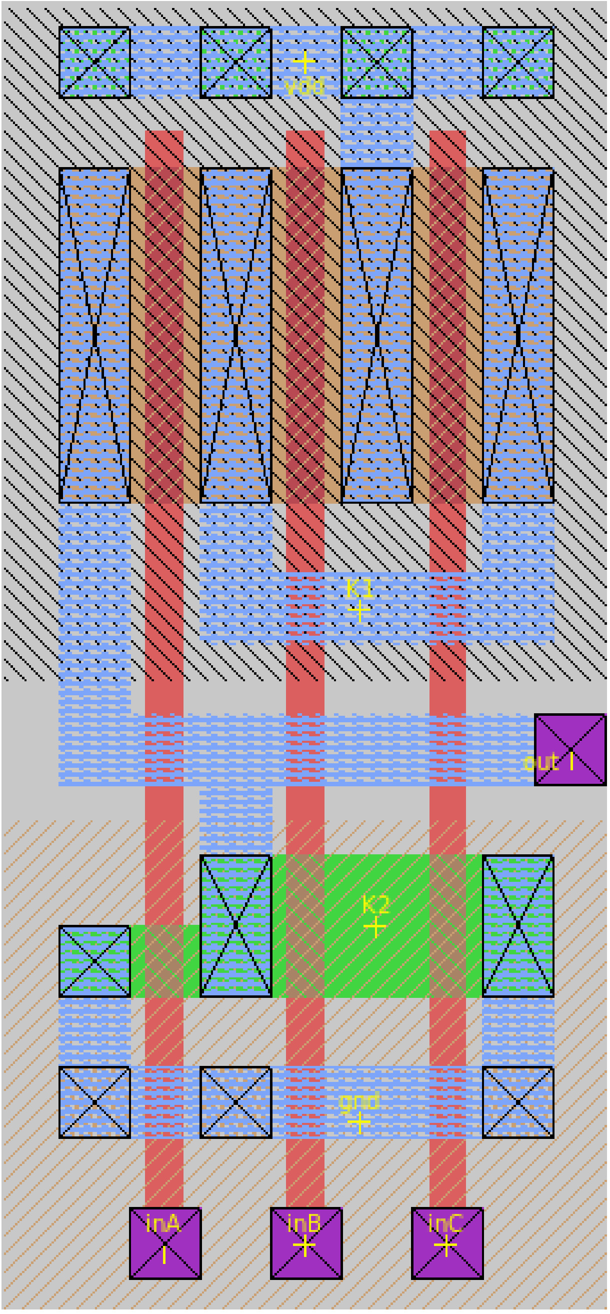


Figure 9: Standard cells implementation of $f_{aoi21}$ binary function using VLSI Magic software.

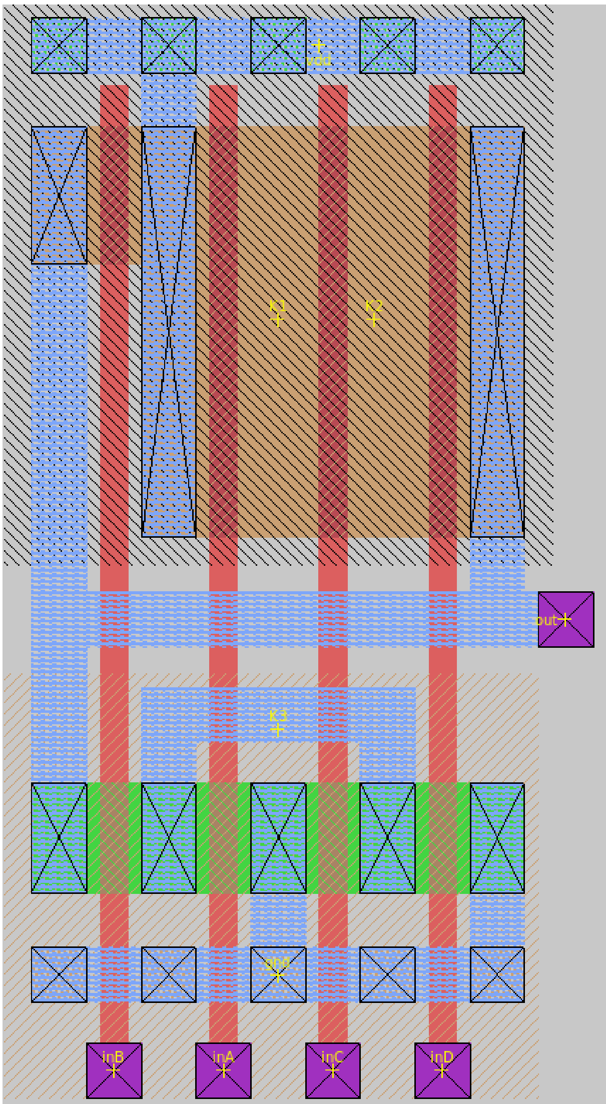


Figure 10: Standard cells implementation of $f_{oai31}$ binary function using VLSI Magic software.

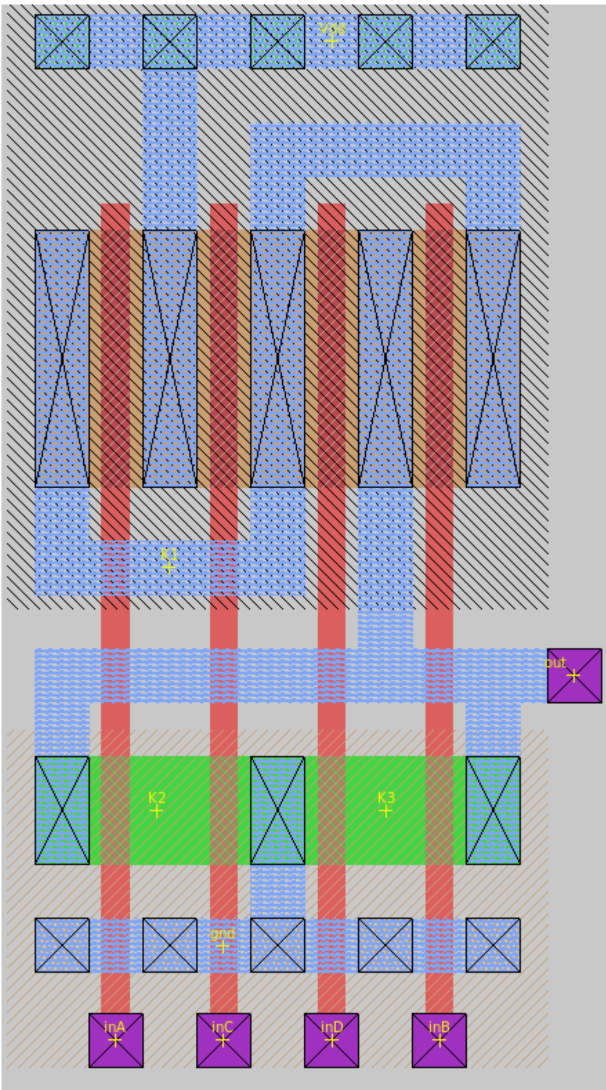


Figure 11: Standard cells implementation of $f_{aoi22}$ binary function using VLSI Magic software.

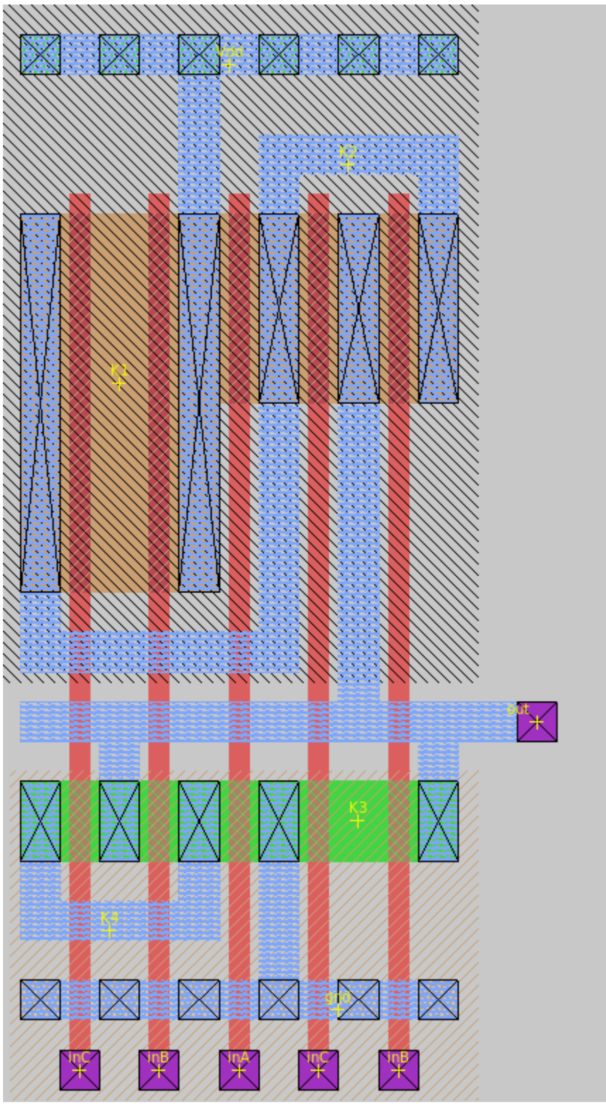


Figure 12: Standard cells implementation of $f_{maj}$ binary function using VLSI Magic software.

In order to verify that the standard cell layout for each Boolean function was designed correctly, the layout structures are extracted into SPICE netlists to perform DC and transient analyses. The physical implementation is validated as correct only if the simulation output responses in SPICE accurately match the theoretical truth tables of each function, as presented in Table 1.

Table 1: Truth Tables for the Implemented 3-Input and 4-Input CMOS Boolean Functions.

(a) 3-Input Functions

| **a** | **b** | **c** | $f_{aoi21}$ | $f_{maj}$ |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 |

(b) 4-Input Functions

| **a** | **b** | **c** | **d** | $f_{oai31}$ | $f_{aoi22}$ |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 | 0 |
| 1 | 0 | 1 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 |

Using the ngspice simulator, we examined the output behavior of each Boolean function across all possible input combinations. The simulation results demonstrated that, for every input stimulus, the transient responses obtained from SPICE perfectly matched the theoretical values of the truth tables. This functional agreement verifies the correctness of the physical layout implementation generated within the Magic VLSI software. The resulting ngspice simulation waveforms for all binary functions are presented in Figures **??**,**??**,**??**,**??**.

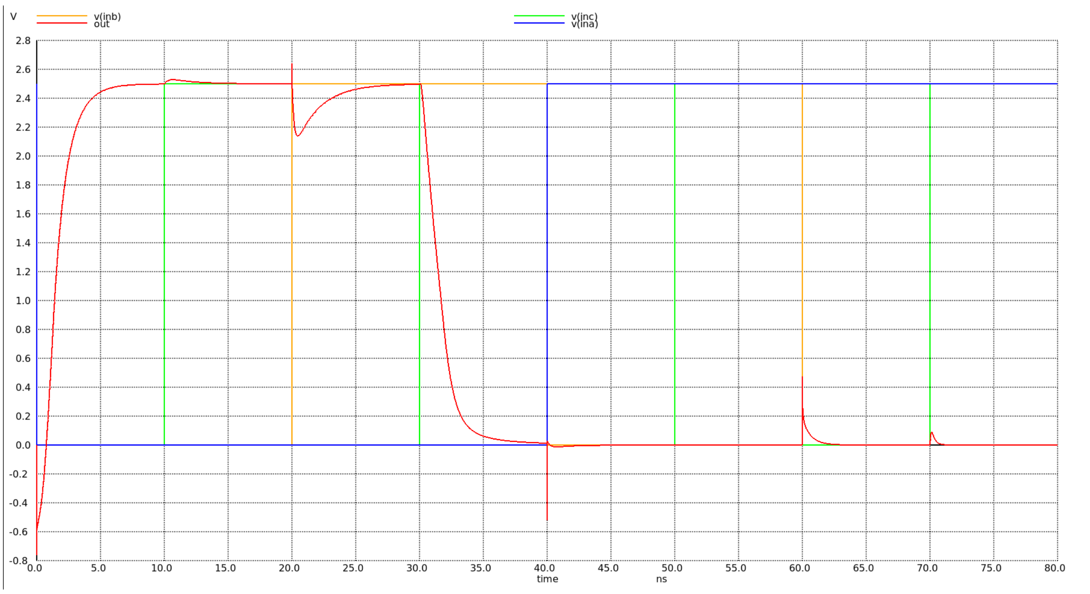


Figure 13: Output voltage levels of $f_{aoi21}$ (in red) for every combination of inputs a,b,c.

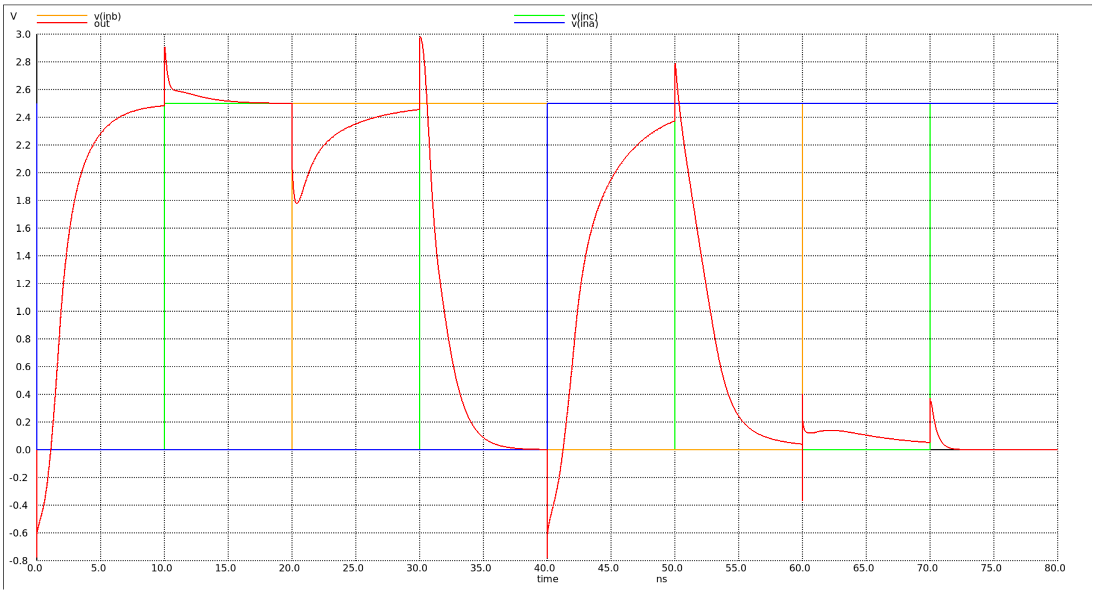


Figure 14: Output voltage levels of $f_{maj}$ (in red) for every combination of inputs a,b,c.

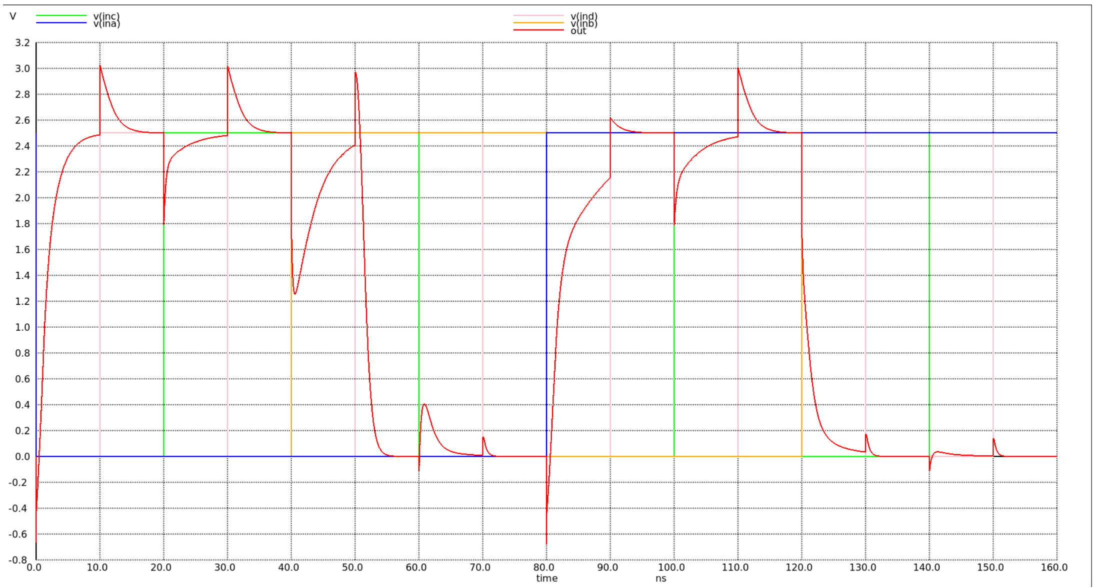


Figure 15: Output voltage levels of $f_{oai31}$ (in red) for every combination of inputs a,b,c,d.

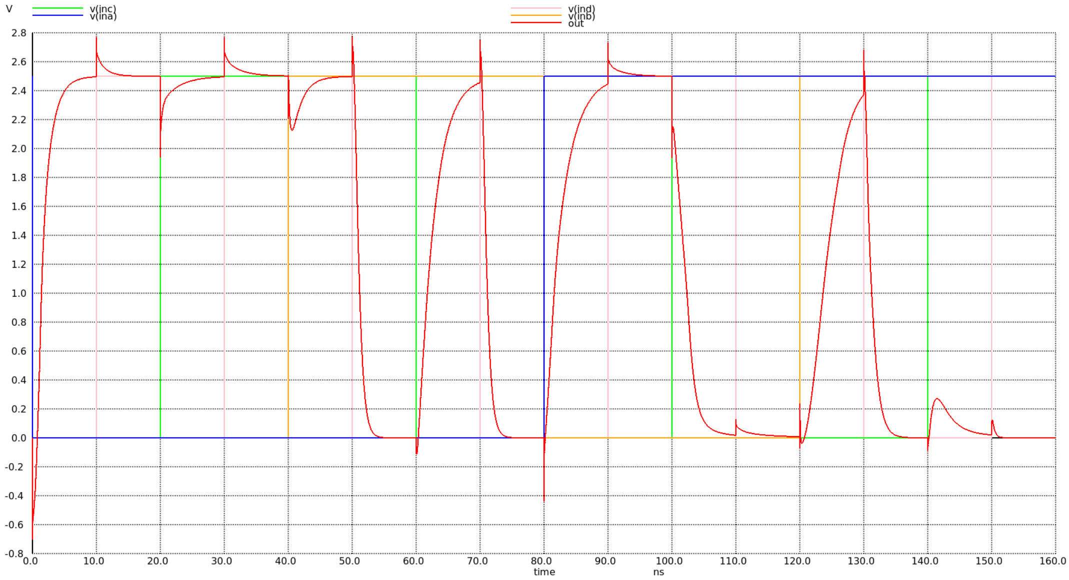


Figure 16: Output voltage levels of $f_{aoi22}$ (in red) for every combination of inputs a,b,c,d.

# 4 Transistor Geometry and Parasitic Extraction

In this section ,based on the physical layout implementation of the $f_{aoi21}$ gate in Magic VLSI 17, the area and perimeter of the source and drain regions for each transistor are calculated. This geometric extraction is performed by counting the respective lambda-based ($\lambda$) layout grid squares for each diffusion area.

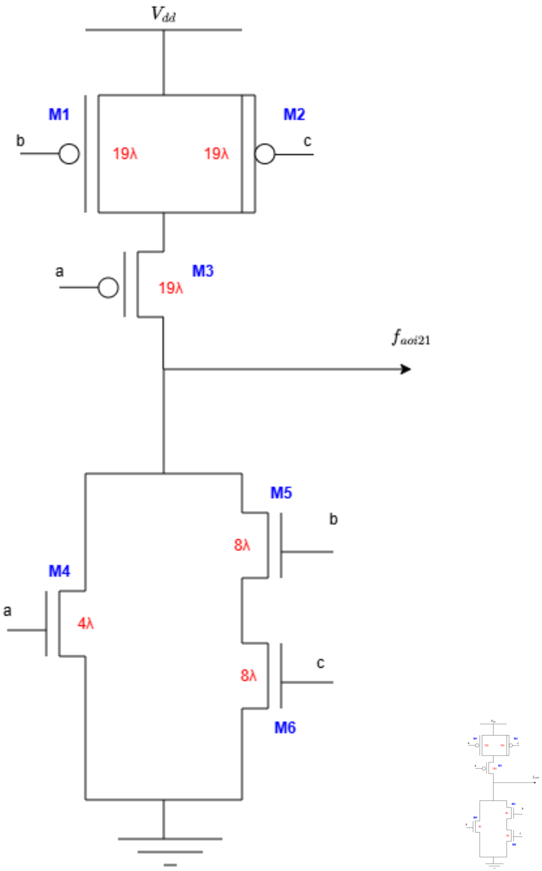


Figure 17: $f_{aoi21}$ function implementation.

These measurements are essential for accurate parasitic capacitance estimation during SPICE simulations. The extracted geometric parameters—including channel width ($W$), the source area ($AS$), the drain area ($AD$), the source perimeter ($PS$) and the drain perimeter ($PD$)—are summarized in Table 2.

Table 2: Extracted Geometric Parameters for the $f_{aoi21}$ Transistors ($\lambda = 0.125\,\mu$m).

| **Transistor** | $W$ **($\mu$m)** | $AS$ **($\mu$m$^2$)** | $AD$ **($\mu$m$^2$)** | $PS$ **($\mu$m)** | $PD$ **($\mu$m)** |
|---|---|---|---|---|---|
| M1 | $19\lambda = 2.375$ | $57\lambda^2 = 0.890$ | $57\lambda^2 = 0.890$ | $6\lambda = 0.75$ | $6\lambda = 0.75$ |
| M2 | $19\lambda = 2.375$ | $57\lambda^2 = 0.890$ | $95\lambda^2 = 1.480$ | $6\lambda = 0.75$ | $29\lambda = 3.625$ |
| M3 | $19\lambda = 2.375$ | $57\lambda^2 = 0.890$ | $95\lambda^2 = 1.480$ | $6\lambda = 0.75$ | $29\lambda = 3.625$ |
| M4 | $4\lambda = 0.500$ | $20\lambda^2 = 0.310$ | $20\lambda^2 = 0.310$ | $14\lambda = 1.75$ | $10\lambda = 1.25$ |
| M5 | $8\lambda = 1.000$ | $24\lambda^2 = 0.375$ | $24\lambda^2 = 0.375$ | $6\lambda = 0.75$ | $6\lambda = 0.75$ |
| M6 | $8\lambda = 1.000$ | $40\lambda^2 = 0.625$ | $24\lambda^2 = 0.375$ | $18\lambda = 2.25$ | $6\lambda = 0.75$ |

Using the geometric parameters extracted from the layout, the diffusion (junction) capacitances for each transistor are calculated. This analysis targets the high-to-low transition ($1 \rightarrow 0$) at the output node. The calculations are based on the specific technological parameters provided for the SCMOS process, which are defined as follows:

- Zero-bias junction capacitance per unit area: $C_j = 2\,\text{fF}/\mu\text{m}^2$
- Zero-bias sidewall junction capacitance per unit perimeter: $C_{jsw} = 0.28\,\text{fF}/\mu\text{m}$
- Grading coefficients for NMOS: $K_{nj} = 0.57$, $K_{njsw} = 0.61$
- Grading coefficients for PMOS: $K_{pj} = 0.79$, $K_{pjsw} = 0.86$

The total diffusion capacitance for any given transistor is governed by the standard expression:

$$C_{db} = K_j \cdot AD \cdot C_j + K_{jsw} \cdot PD \cdot C_{jsw} \tag{7}$$

By substituting the extracted layout dimensions and the technology constants above, the individual parasitic components for the $f_{aoi21}$ gate are derived as follows:

$$C_{db1} = 0.79 \cdot 0.89 \cdot 2 + 0.86 \cdot 0.75 \cdot 0.28 = 1.588\,\text{fF} \tag{8}$$
$$C_{db2} = 0.79 \cdot 1.48 \cdot 2 + 0.86 \cdot 3.625 \cdot 0.28 = 3.210\,\text{fF} \tag{9}$$
$$C_{db3} = 0.79 \cdot 1.48 \cdot 2 + 0.86 \cdot 3.625 \cdot 0.28 = 3.210\,\text{fF} \tag{10}$$
$$C_{db4} = 0.57 \cdot 0.31 \cdot 2 + 0.61 \cdot 1.25 \cdot 0.28 = 0.560\,\text{fF} \tag{11}$$
$$C_{db5} = 0.57 \cdot 0.375 \cdot 2 + 0.61 \cdot 0.75 \cdot 0.28 = 0.550\,\text{fF} \tag{12}$$
$$C_{db6} = 0.57 \cdot 0.375 \cdot 2 + 0.61 \cdot 0.75 \cdot 0.28 = 0.550\,\text{fF} \tag{13}$$

# 5 Fan-Out Configuration and Load Analysis

We now assume a fan-out configuration where the output of the initial $f_{aoi21}$ circuit implementation directly drives the inputs of four identical downstream stages. More specifically, the output of the primary $f_{aoi21}$ gate is connected to input $b$ of both the Pull-Up and Pull-Down networks across all four subsequent implementations. This configuration loads the initial circuit's output node with a total of eight transistor gates, as schematically illustrated in Figure 18.

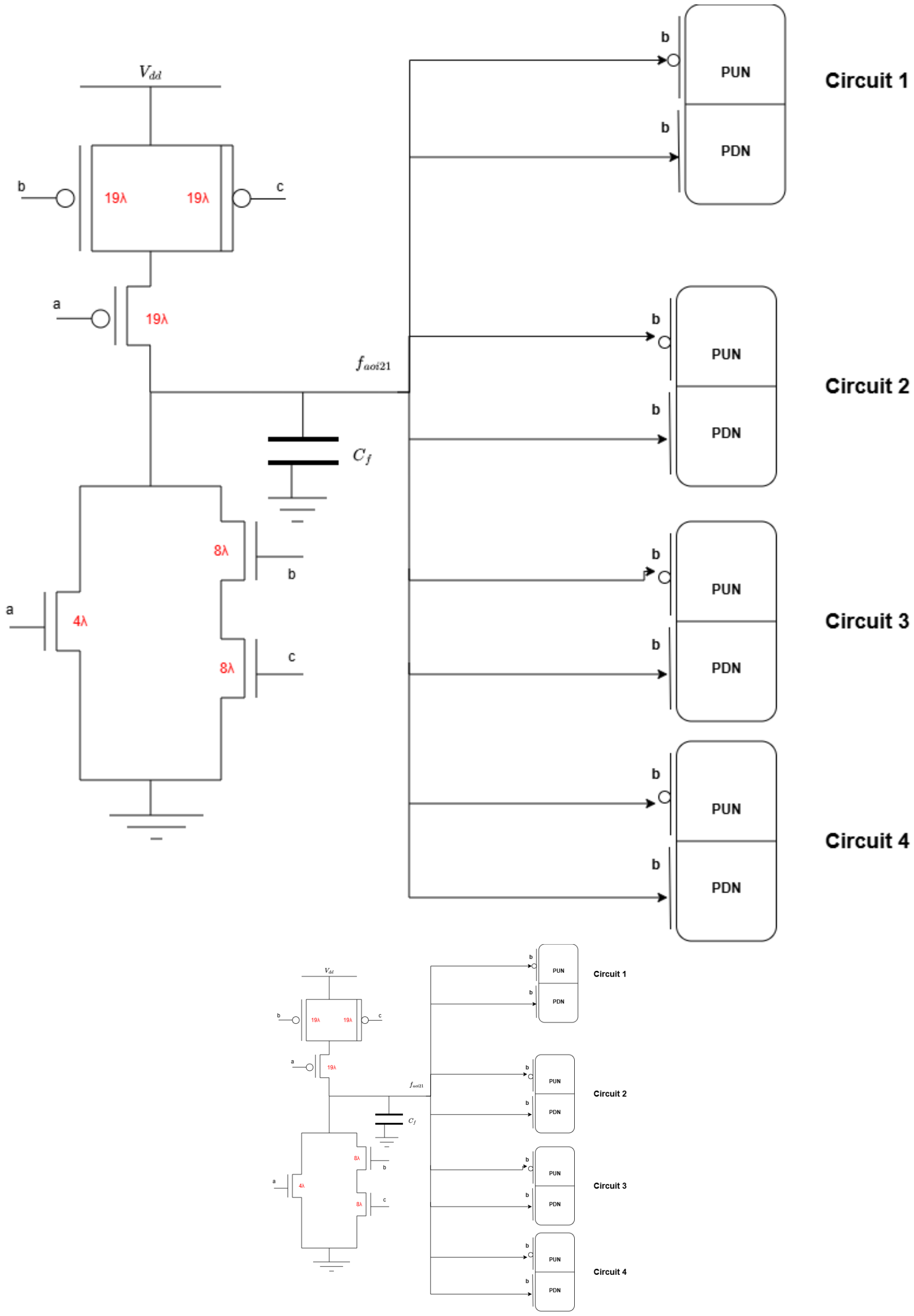


Figure 18: Schematic representation of the Fan-Out 4 (FO4) loading configuration, where the output of the primary $f_{aoi21}$ gate drives the $b$ inputs (both PMOS and NMOS gates) of four identical downstream circuits.

# 6 Analytical Propagation Delay Estimation using Elmore Metric

## 6.1 Worst-Case Delay Path and Elmore Formulation

For the evaluation of the Pull-Down Network (PDN) of the $f_{aoi21}$ standard cell, the discharge path involving the two series-connected NMOS transistors is selected to determine the worst-case propagation delay ($t_{pHL}$). The total capacitance analyzed along this critical path includes the internal node capacitance, $C_{bc}$, located at the junction between the transistors controlled by inputs $b$ and $c$, and the output interface capacitance, $C_f$, where the Pull-Up Network (PUN) and PDN interconnect.

Since the objective is to isolate and evaluate the worst-case delay of the PDN exclusively during a high-to-low transition, the PUN does not actively contribute to the switching dynamics and is treated as a parasitic load at the output node. Finally, the external loading effects—specifically the gate capacitances of the four downstream transistor pairs driven by this output (Fan-Out 4 configuration)—are integrated into the cumulative load capacitance calculation.

Based on the equivalent RC network of the Pull-Down Network, the total propagation delay is mathematically modeled using the Elmore delay metric and is governed by the following expression:

$$\tau_{RC} = \ln(2) \cdot [C_{bc} \cdot R_1 + C_L \cdot (R_1 + R_2)] \tag{14}$$

Where:

- $C_{bc}$ represents the internal parasitic node capacitance between the series NMOS devices.
- $C_L$ represents the total effective load capacitance at the output node.
- $R_1, R_2$ denote the equivalent channel resistances of the two series-connected NMOS transistors forming the discharge path.

Given that the cumulative resistance of the worst-case Pull-Down Network path is calculated to be $R_{\text{PDN}} = 6.5\,\text{k}\Omega$, and assuming identical sizing for the series-connected devices, the individual channel resistances are evenly distributed as:

$$R_1 = R_2 = 3.25\,\text{k}\Omega \tag{15}$$

—

## 6.2 Internal Node Capacitance Evaluation ($C_{bc}$)

To achieve a precise Elmore delay estimation, the total parasitic capacitance of the internal storage node, $C_{bc}$, must be quantified. This internal node capacitance is composed of the source junction capacitance of transistor $b$ ($C_{s,b}$), the drain junction capacitance of transistor $c$ ($C_{d,c}$), and the structural gate-to-source ($C_{gs}$) and gate-to-drain ($C_{gd}$) overlap capacitances contributing to the node:

$$C_{bc} = C_{s,b} + C_{d,c} + 2 \cdot C_{gs,b} + 2 \cdot C_{gd,b} \tag{16}$$

By utilizing the extracted layout geometric parameters and the technology parameter $C_{On} = 0.31\,\text{fF}/\mu\text{m}$ for the overlap capacitance components, the individual terms are evaluated as follows:

$$\begin{aligned} C_{s,b} &= K_{nj} \cdot AD_n \cdot C_j + K_{njsw} \cdot PD_n \cdot C_{jsw} \\ &= 0.57 \cdot 0.375 \cdot 2 + 0.61 \cdot 0.75 \cdot 0.28 = 0.550\,\text{fF} \quad (17) \\ C_{d,c} &= K_{nj} \cdot AD_n \cdot C_j + K_{njsw} \cdot PD_n \cdot C_{jsw} \\ &= 0.57 \cdot 0.375 \cdot 2 + 0.61 \cdot 0.75 \cdot 0.28 = 0.550\,\text{fF} \quad (18) \\ 2 \cdot C_{gs,b} &= 2 \cdot C_{On} \cdot W_b = 2 \cdot 0.31 \cdot 1.000 = 0.620\,\text{fF} \quad (19) \\ 2 \cdot C_{gd,b} &= 2 \cdot C_{On} \cdot W_c = 2 \cdot 0.31 \cdot 1.000 = 0.620\,\text{fF} \quad (20) \end{aligned}$$

Summing these individual parasitic elements, the cumulative internal node capacitance is determined to be:

$$C_{bc} = 0.550 + 0.550 + 0.620 + 0.620 = 2.340\,\text{fF} \quad (21)$$

—

## 6.3 Output Node Loading and Final Delay Calculation ($C_L$, $\tau_{RC}$)

First, the parasitic capacitance contribution at the output node from the local Pull-Down Network, denoted as $C_F$, is evaluated. This term accounts for the drain junction capacitance of the switching NMOS device and its corresponding gate-to-drain overlap capacitances:

$$C_F = C_{d,b(\text{nmos})} + 2 \cdot C_{gd,b(\text{nmos})} = 0.550 + 2 \cdot 0.31 \cdot 1.000 = 1.170\,\text{fF} \quad (22)$$

Next, the total effective load capacitance, $C_L$, present at the output node is determined. As established in the Fan-Out 4 scenario, the node is loaded by the combined gate capacitances of 4 downstream PMOS and 4 downstream NMOS transistor stages. The analytical expansion for $C_L$ integrates both the intrinsic oxide capacitance ($C_{ox} \cdot W \cdot L$) and the extrinsic overlap components ($2 \cdot C_{On} \cdot W$):

$$\begin{aligned} C_L &= C_F + 4 \cdot C_{gB(\text{nmos})} + 4 \cdot C_{gB(\text{pmos})} \\ &= 1.170 + 4 \cdot \left[2 \cdot C_{On(\text{nmos})} \cdot W_n + C_{ox} \cdot W_n \cdot L_n\right] + 4 \cdot \left[2 \cdot C_{On(\text{pmos})} \cdot W_p + C_{ox} \cdot W_p \cdot L_p\right] \\ &= 1.170 + 4 \cdot [2 \cdot 0.31 \cdot 1.000 + 6 \cdot 1.000 \cdot 0.25] + 4 \cdot [2 \cdot 0.27 \cdot 2.375 + 6 \cdot 2.375 \cdot 0.25] \\ &= 48.030\,\text{fF} \quad (23) \end{aligned}$$

Finally, by substituting the evaluated capacitance terms ($C_{bc} = 2.340$ fF and $C_L = 48.030$ fF) and the distributed channel resistances ($R_1 = R_2 = 3.25\,\text{k}\Omega$) into the Elmore delay metric, the worst-case propagation delay ($\tau_{RC}$) is derived:

$$\begin{aligned} \tau_{RC} &= \ln(2) \cdot [C_{bc} \cdot R_1 + C_L \cdot (R_1 + R_2)] \\ &= \ln(2) \cdot [2.340\,\text{fF} \cdot 3.25\,\text{k}\Omega + 48.030\,\text{fF} \cdot (3.25\,\text{k}\Omega + 3.25\,\text{k}\Omega)] \\ &= 221.660\,\text{ps} \quad (24) \end{aligned}$$

Consequently, the theoretical worst-case high-to-low propagation delay of the designed $f_{aoi21}$ standard cell under a heavy Fan-Out 4 loading condition is established at approximately **221.660 ps**.